MODULE III

Differential Calculus

Functions of Two Variables

A real-valued **function of two variables** f, consists of

1. A set A of **ordered pairs** of real numbers (x, y) called the **domain** of the function.
2. A **rule** that associates with each ordered pair in the domain of f **one and only one** real number, denoted by $z = f(x, y)$.

$$d(uv) = u\,d(v) + v\,d(u)$$ Product Rule

$$d\left(\frac{u}{v}\right) = \frac{v\,d(u) - u\,d(v)}{v^2}$$ Quotient Rule

$z = f(x, y)$

$z = x^2y$

$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

$\frac{\partial z}{\partial x} = 2xy$

$\frac{\partial z}{\partial x} = f_x$

$\frac{\partial z}{\partial y} = f_y$

$$z = f(x,y) = \boxed{x \log x + y^2}$$

$$\frac{\partial z}{\partial x} = f_x = \left\{ x \cdot \frac{1}{x} + \log x \cdot 1 \right\} + y^2$$

$$\frac{\partial z}{\partial y} = f_y = 2y$$

$$z = 2e^{x^2} y^2$$

$$\frac{\partial z}{\partial x} = f_x = 4e^{x^2} y^2$$

$$= 2e^{x^2}(2x) \cdot y^2$$

$$= 4xe^{x^2} y^2$$

$$d(x^n) = nx^{n-1}$$

$$= 2x^1$$

$$d(e^x) = e^x \cdot 1$$

$$= e^x$$

$$d(e^{x^2}) = e^{x^2}$$

$$= e^{x^2} \cdot 2x$$

$$d(e^x) = e^x \cdot 1$$

$$d(e^{x^2}) = e^{x^2}$$

$$= e^{x^2} \cdot 2x$$

$$= 1.$$

$$\frac{\partial z}{\partial y} = f_y = 2e^{x^2} \cdot 2y = 4e^{x^2} y$$

$$2e^{x^2} y$$

Examples

- **Let *f* be the function defined by**

$$f(x,y) = x + xy + y^2 + 2$$

- **Compute *f*(0, 0), *f*(1, 2), and *f*(2, 1).**

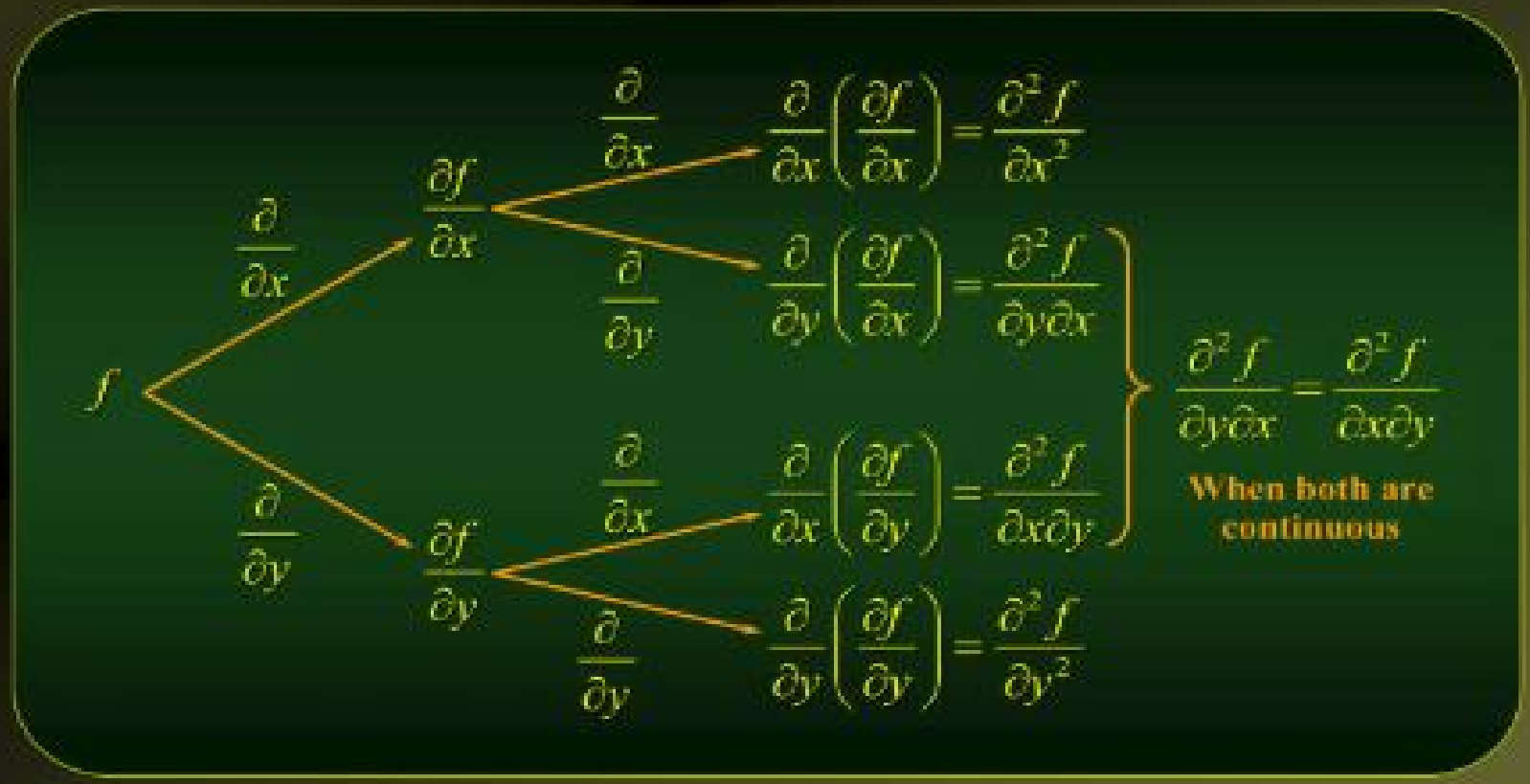
Solution

$$f(0,0) = 0 + (0)(0) + 0^2 + 2 = 2$$

$$f(1,2) = 1 + (1)(2) + 2^2 + 2 = 9$$

$$f(2,1) = 2 + (2)(1) + 1^2 + 2 = 7$$

Partial Derivatives



First Partial Derivatives

First Partial Derivatives of $f(x, y)$

- Suppose $f(x, y)$ is a function of two variables x and y.
- Then, the first partial derivative of f with respect to x at the point (x, y) is

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

provided the limit exists.

- The first partial derivative of f with respect to y at the point (x, y) is

$$\frac{\partial f}{\partial y} = \lim_{k \to 0} \frac{f(x, y+k) - f(x, y)}{k}$$

Examples

- Find the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ of the function

$$f(x,y) = x^2 - xy^2 + y^3$$

$f(x,y) = x^2 - xy^2 + y^3$

$\frac{\partial f}{\partial x} = f_x = 2x - y^2 + 0$

$\frac{\partial f}{\partial y} = f_y = -2xy + 3y^2$

$d(x^n) = n x^{n-1}$

- Find the **partial derivatives** $\partial f/\partial x$ and $\partial f/\partial y$ of the **function**

$$f(x, y) = x^2 - xy^2 + y^3$$

- Use the partials to determine the **rate of change** of f in the x-**direction** and in the y-**direction** at the point $(1, 2)$.

$f_x = 2x - y^2$

$f_x(1,2) = 2(1) - (2)^2 = -2$

$f_y(1,2) = 8$

$f_y = -2xy + 3y^2$

Examples

- Find the first partial derivatives of the function

$$g(s,t) = (s^2 - st + t^2)^5$$

- Find the first partial derivatives of the function

$$w(x, y) = \frac{xy}{x^2 + y^2}$$

- Find the first partial derivatives of the function

$$h(u,v) = e^{u^2 - v^2}$$

- Find the first partial derivatives of the function

$$w = f(x, y, z) = xyz - xe^{yz} + x \ln y$$

▷ $g(s,t) = (s^2 - st + t^2)^5$ $\Big| = (u)^5$

$\frac{\partial g}{\partial s} = 5(s^2 - st + t^2)^4 (2s - t)$ $\Big| = 5u^4 (du)$

$\frac{\partial g}{\partial t} = 5(s^2 - st + t^2)^4 (-s + 2t)$ $\Big|$

2) $w(x,y) = \frac{xy}{x^2+y^2}$ $\quad \because d\left(\frac{u}{v}\right) = \frac{v\,d(u) - u\,d(v)}{v^2}$

$$\frac{\partial w}{\partial x} = \frac{(x^2+y^2)y - (xy)\cdot 2x}{(x^2+y^2)^2}$$

$$= \frac{y(x^2+y^2) - 2x^2y}{(x^2+y^2)^2} = \frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{y(x^2+y^2)-2x^2y}{(x^2+y^2)^2}$$

$$= \frac{x^2y+y^3-2x^2y}{(x^2+y^2)^2} = \frac{y^3-x^2y}{(x^2+y^2)^2}$$

$$=$$

$$\frac{\partial w}{\partial y} = \frac{(x^2+y^2)x - xy(2y)}{(x^2+y^2)^2}$$

$$= \frac{x^3+y^2x-2xy^2}{(x^2+y^2)^2}$$

$$= \frac{x^3-xy^2}{(x^2+y^2)^2} \quad //$$

3) $h(u,v) = e^{u^2-v^2}$

$\therefore d(e^x) = e^x d(x)$

$= e^x$

$$\frac{\partial h}{\partial u} = e^{u^2-v^2}(2u)$$

$$\frac{\partial h}{\partial v} = e^{u^2-v^2}(-2v) \;//$$

4) $f(x,y,z) = xyz - xe^{yz} + x\log y$

$\frac{\partial f}{\partial x} = yz - e^{yz} + \log y$

$\frac{\partial f}{\partial y} = xz - xze^{yz} + \frac{x}{y} - \frac{1}{y}$

$\frac{\partial f}{\partial z} = xy - xye^{yz} + 0$

Second Order Partial Derivatives

- Differentiating the function f_x with respect to x leads to the second partial derivative

$$f_{xx} \equiv \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}(f_x)$$

- But the function f_x can also be differentiated with respect to y leading to a different second partial derivative

$$f_{xy} \equiv \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}(f_x)$$

Second Order Partial Derivatives

- Similarly, differentiating the function f_y with respect to y leads to the second partial derivative

$$f_{yy} \equiv \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(f_y)$$

- Finally, the function f_y can also be differentiated with respect to x leading to the second partial derivative

$$f_{yx} \equiv \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(f_y)$$

Examples

- **Find the second-order partial derivatives of the function**

$$f(x,y) = x^3 - 3x^2y + 3xy^2 + y^2$$

- **Find the second-order partial derivatives of the function**

$$f(x,y) = e^{xy^2}$$

$$f(x,y) = x^3 - 3x^2y + 3xy^2 + y^2$$

$$\frac{\partial f}{\partial x} = 3x^2 - 6xy + 3y^2$$

$$\frac{\partial f}{\partial y} = -3x^2 + 6xy + 2y$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(3x^2 - 6xy + 3y^2\right)$$

$$= 6x - 6y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(-3x^2+6xy+2y\right)$$

$$= 6x+2$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(-3x^2+6xy+2y\right)$$

$$= (-6x+6y)$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = -6x+6y$$

THE CHAIN RULE

- If $y = f(x)$ and $x = g(t)$, where f and g are differentiable functions, then y is indirectly a differentiable function of t, and

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

$$y = f(x), \quad x = g(s,t)$$

$$\frac{\partial y}{\partial s} = \frac{\partial y}{\partial x}\frac{\partial x}{\partial s}$$

$$\frac{\partial y}{\partial t} = \frac{\partial y}{\partial x}\frac{\partial x}{\partial t}$$

THE CHAIN RULE (CASE 1)

- Suppose that $z = f(x, y)$ is a differentiable function of x and y, where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t.

  – Then, z is a differentiable function of t and

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}$$

THE CHAIN RULE

- For functions of more than one variable, the Chain Rule has several versions.

THE CHAIN RULE (CASE 2)

- Suppose $z = f(x, y)$ is a differentiable function x and y, where $x = g(s, t)$ and $y = h(s, t)$ are differentiable functions of s and t.

  — Then,

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

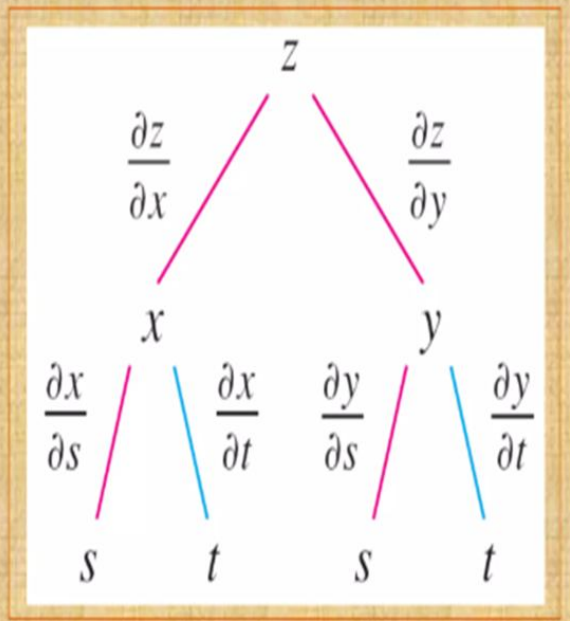
THE CHAIN RULE

- Case 2 of the Chain Rule contains three of variables:
  - s and t are independent variables.
  - x and y are called intermediate variables.
  - z is the dependent variable.

TREE DIAGRAM

- Then, we draw branches from x and y to the independent variables s and t.

– On each branch, we write the corresponding partial derivative.



$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

If $z = e^x \sin y$, where $x = st^2$ and $y = s^2t$, find $\partial z/\partial s$ and $\partial z/\partial t$.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= e^x \sin y \cdot t^2 + e^x \cos y \cdot 2st$$

$$= e^x t(t \sin y + 2s \cos y) //$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$\frac{\partial z}{\partial t} = e^x \sin y \cdot 2st + e^x \cos y \cdot s^2 \cdot 1$$

$$= e^x s\,(2t \sin y + s \cos y)$$

$z = e^{xy}$ $\quad x = 2u + v \quad y = \frac{u}{v}$

$u \cdot \frac{1}{v}$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$= e^{xy} \cdot y \cdot 2 + e^{xy} \cdot x \cdot \frac{1}{v}$$

$$= e^{xy}\left(2y + \frac{x}{v}\right)$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

$y = \frac{u}{v}$

$$= y e^{xy} \cdot 1 + x e^{xy}\left(u\left(\frac{-1}{v^2}\right)\right) = u.$$

$$= e^{xy}\left(y - \frac{xu}{v^2}\right) //$$

$$w = \sqrt{x^2 + y^2 + z^2} \text{ and } x = \cos t, y = \sin t,$$
$$z = \tan t \text{ and } t = \frac{\pi}{4}$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= \frac{1}{2\sqrt{x^2+y^2+z^2}}(2x)(-\sin t) + \frac{1}{2\sqrt{x^2+y^2+z^2}}(2y)(\cos t)$$
$$+ \frac{1}{2\sqrt{x^2+y^2+z^2}}(2z)(\sec^2 t)$$

$$= \frac{1}{\sqrt{x^2+y^2+z^2}}\left(-x\sin t + y\cos t + z\sec^2 t\right)$$

$$\text{At } t = \frac{\pi}{4} = 45^\circ$$

$$= \frac{1}{\sqrt{x^2+y^2+z^2}}\left(-x\left(\frac{1}{\sqrt{2}}\right) + y\frac{1}{\sqrt{2}} + z(\sqrt{2})^2\right)$$

$$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{1/\sqrt{2}} = \sqrt{2}$$

$$= \frac{1}{\sqrt{x^2+y^2+z^2}} \left(\frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}} + 2z \right) //$$

$$z = f(x, y) = 4x^2 + 3y^2$$
$$x(t) = \sin t$$
$$y(t) = \cos t$$

$$z = f(x, y) = 3x^2 - 2xy + y^2$$
$$x(u, v) = 3u + 2v$$
$$y(u, v) = 4u - v.$$

Use the Chain Rule to find the indicated partial derivatives.

$z = x^4 + xy^3, \quad x = uv^4 + w^3, \quad y = u + ve^w$

$\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}, \frac{\partial z}{\partial w}$ when $u = 1, \ v = 1, \ w = 0$

$z = xy + y \quad x = \cos\theta \quad y = \sin\theta$

$\frac{dz}{d\theta} = ?$ when $\theta = \frac{\pi}{2}$

Method of finding critical points

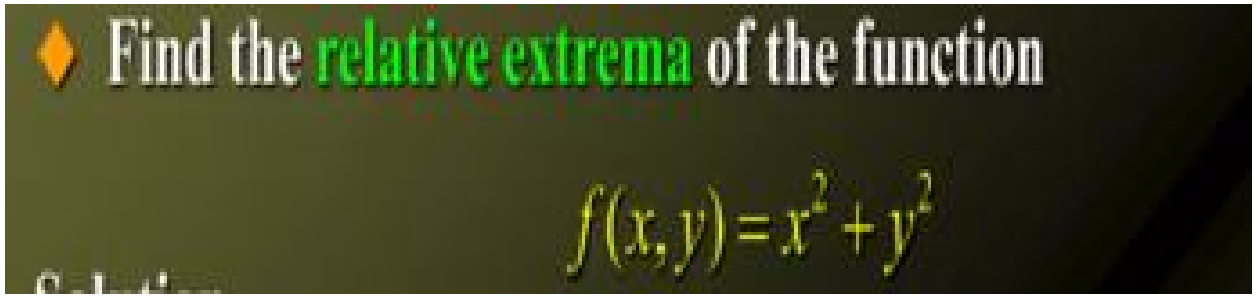
1. **Find the critical points of $f(x, y)$ by solving the system of simultaneous equations**

$$f_x = 0 \qquad f_y = 0$$

2. **The second derivative test: Let**

$$D(x, y) = f_{xx} f_{yy} - f^2{}_{xy}$$

3. **Then,**
   a. $D(a, b) > 0$ **and** $f_{xx}(a, b) < 0$ **implies that** $f(x, y)$ **has a relative maximum at the point** (a, b).
   b. $D(a, b) > 0$ **and** $f_{xx}(a, b) > 0$ **implies that** $f(x, y)$ **has a relative minimum at the point** (a, b).
   c. $D(a, b) < 0$ **implies that** $f(x, y)$ **has neither a relative maximum nor a relative minimum at the point** (a, b), **it has instead a saddle point.**
   d. $D(a, b) = 0$ **implies that the test is inconclusive, so some other technique must be used to solve the problem.**



$f(x,y) = x^2 + y^2$

$f_x = \frac{\partial f}{\partial x} = 2x = 0 \Rightarrow x = 0$

$f_y = \frac{\partial f}{\partial y} = 2y = 0 \Rightarrow y = 0$

$\therefore$ critical point is $(x,y) = (0,0)$

$$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(2x) = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(2y) = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(2y) = 2$$

$$\therefore D = rt - s^2 = (2)(2) - (0)^2 = 4 > 0$$

$D = 4 > 0$, $\frac{\partial^2 f}{\partial x^2} = f_{xx} = 2 > 0$

$\therefore f(x,y)$ has minimum value

$\therefore f(x,y) = x^2 + y^2$

$\therefore f(0,0) = (0)^2 + (0)^2 = 0$ is the minimum value.

$$f(x,y)=3x^2-4xy+4y^2-4x+8y+4$$

$$f_x=\frac{\partial f}{\partial x}=6x-4y-4=0$$

$$f_y=\frac{\partial f}{\partial y}=-4x+8y+8=0$$

$\therefore$ $6x-4y-4=0 \times 2$

$-4x+8y+8=0 \times 1$

$$12x - 8y - 8 = 0$$
$$+\quad -4x + 8y + 8 = 0$$

$$8x = 0$$

$$\boxed{x = 0},$$

$$-4x + 8y + 8 = 0$$
$$0 + 8y + 8 = 0$$
$$8y = -8$$
$$\boxed{y = -1}$$

$\therefore$ Critical point is $(0, -1)$

$$r = \frac{\partial^2 f}{\partial x^2} = 6$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -4$$

$$t = \frac{\partial^2 f}{\partial y^2} = 8$$

$$\therefore D = rt - s^2 = (6)(8) - (-4)^2$$

$$= 48 - 16 = 32 > 0$$

$\therefore D = 32 > 0, \quad f_{xx} = 6 > 0$

$\therefore f(x,y)$ has minimum value

$f(x,y) = 3x^2 - 4xy + 4y^2 - 4x + 8y + 4$

$f(0,-1) = 0$ is the minimum value

- Find the relative extrema of the function

$$f(x,y) = 4y^3 + x^2 - 12y^2 - 36y + 2$$

$f_x = 2x = 0 \Rightarrow x = 0$

$f_y = 12y^2 - 24y - 36 = 0 \Rightarrow y =$

$y^2 - 2y - 3 = 0 \Rightarrow y = -1, 3$

$\therefore$ (critical points are $(0,-1)$ & $(0,3)$)

$r = f_{xx} = 2$

$s = f_{xy} = 0$

$t = f_{yy} = 24y - 24 = 24(y-1)$

$\therefore D = rt - s^2 = 2 \cdot 24(y-1) - (0)$

$= 48(y-1)$

$=$

$y-1 > 0$
$y > 1$
$y-1 < 0$
$y < 1$

$y-1 > 0$
$y > 1$
$y-1 < 0$

$D = 48(y-1)$

$D(0,-1) = 48(-1-1) = -2\times 48 = -96 < 0$

$\therefore f(x,y)$ has a saddle point

$D(0,3) = 48(3-1) = 48\times 2 = 96 > 0$

$f_{xx} = 2 > 0$

$f(x,y)$ has minimum value

$\therefore f(x,y) = 4y^3 + x^2 - 12y^2 - 36y + 2$

$f(0,3) = -106$ is the minimum value

Find the critical points of the function:

$$f(x, y) = x^3 - xy + y^2$$

Given $f(x, y) = -3 + 7x + 7x^2 + 3y + 6y^2$, find the find the critical points. Determine if there is a minimum, maximum, or saddle point at that location.

Lagrange Multipliers

Method of Lagrange Multipliers

A method to find the local minimum and maximum of a function with two variables subject to conditions or constraints on the variables involved.

Suppose that, subject to the constraint $g(x,y)=0$, the function $z=f(x,y)$ has a local maximum or a local minimum at the point

$$(x_0, y_0)$$

Form the function

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

Then there is a value of λ such that (x_0, y_0, λ) is a solu of the system of equations

$$\frac{\partial F}{\partial x} = \frac{\partial f}{\partial x} + \lambda \frac{\partial g}{\partial x} = 0 \quad \cdots(1)$$

$$\frac{\partial F}{\partial y} = \frac{\partial f}{\partial y} + \lambda \frac{\partial g}{\partial y} = 0 \quad \cdots(2)$$

$$\frac{\partial F}{\partial \lambda} = g(x, y) = 0 \quad \cdots(3)$$

provided all the partial derivatives exists.

The Problem: Find the extrema of $f(x, y)$ subject to a constraint equation $g(x, y) = k$. Slideshare +1

Lagrange's Theorem: At the optimal constrained point, the gradient vectors are parallel:

$$\nabla f = \lambda \nabla g$$

System of Equations to Solve:

1. $f_x = \lambda g_x$
2. $f_y = \lambda g_y$
3. $g(x, y) = k$

Where λ is the **Lagrange multiplier** (represents the sensitivity of the objective function to changes in the constraint value k).

Steps for Using the Method of Lagrange Multipliers

Step 1: Write the function to be maximized (or minimized) and the constraint in the form:

Find the maximum (or minimum) value of

$$z = f(x, y)$$

subject to the constraint $g(x, y) = 0$

Step 2: Construct the function F:

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

Step 3: Set up the system of equations

$$\frac{\partial F}{\partial x} = 0 \quad \cdots(1)$$

$$\frac{\partial F}{\partial y} = 0 \quad \cdots(2)$$

$$\frac{\partial F}{\partial \lambda} = g(x, y) = 0 \cdots(3)$$

Step 4: Solve the system of equations for x, y and λ.

Step 5: Test the solution (x_0, y_0, λ) to determine maximum or minimum point.

$D = rt - s^2$

Find $D^* = Fxx \cdot Fyy - (Fxy)^2$

If $D^* > 0 \Rightarrow$
- $Fxx < 0$ $\therefore$ maximum point
- $Fxx > 0$ $\therefore$ minimum point

$D^* \leq 0 \Rightarrow$ Test is inconclusive

Step 6: Evaluate $z = f(x, y)$ at each solution (x_0, y_0, λ) found in Step 5.

Find the minimum of

$$f(x,y) = 5x^2 + 6y^2 - xy$$

subject to the constraint

$$x+2y = 24$$

$\nabla f = \lambda \nabla g$

$\nabla f - \lambda \nabla g$

$f(x,y) = 5x^2 + 6y^2 - xy$

$g(x,y) = x + 2y - 24$ $\quad \therefore x+2y = 24$

$F(x,y,\lambda) = f(x,y) + \lambda g(x,y)$

$= (5x^2 + 6y^2 - xy) + \lambda(x+2y-24)$

$$\frac{\partial f}{\partial x} = (10x - y) + \lambda(1) = 0$$

$$= 10x - y = -\lambda \quad \text{①}$$

$$y - 10x = \lambda \quad \text{①}$$

$$\frac{\partial f}{\partial y} = (12y - x) + \lambda(2) = 0$$

$$12y - x = -2\lambda \quad \text{②}$$

$$x - 12y = 2\lambda$$

$$\frac{\partial f}{\partial \lambda} = g(x, y) = x + 2y - 24 = 0 \quad \text{③}$$

put eq² ① in eq² ②

$$x - 12y = 2(y - 10x)$$

$$x - 12y = 2y - 20x$$

$$x + 20x = 2y + 12y$$

$$21x = 14y$$

$$y = \frac{21}{14}x = \frac{3}{2}x$$

$$\therefore y = \frac{3}{2}x$$

∴ put $y = \frac{3}{2}x$ in the constraint $x + 2y - 24 = 0$

$$x + 2\left(\frac{3}{2}x\right) - 24 = 0$$

$$x + 3x = 24$$

$$4x = 24$$

$$\boxed{x = 6}$$

$$\therefore y = 9$$

$$\lambda = -51$$

$f_{xx} = +10 = r$

$f_{xy} = -1 = s$

$f_{yy} = +12 = t$

$D = f_{xx} f_{yy} - s^2$

$= (+10)(+12) - (-1)^2$

$= 120 > 0$

$f_{xx} = +10 > 0.$

$\therefore f(x,y,\lambda)$ has ~~max~~ min Value

$f(x,y,\lambda) = (5x^2+6y^2-xy) + \lambda(x+2y-24)$

$x = 6,\ y = 9,\ \lambda = -51$

$f(6,9,-51) = 612$ is the minimum value

Maximize $z = P(x, y) = x^2 + 3xy - 6y$

Subject to constraint $g(x, y) = x + y - 42 = 0$

Find the minimum of the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $x + y + z = 1$.

Use the method of Lagrange multipliers to find the minimum value of the function

$$f(x, y, z) = x + y + z$$

subject to the constraint $x^2 + y^2 + z^2 = 1$.

Use the method of Lagrange multipliers to find the minimum value of $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x + 2y = 7$.